Pathwise Perturbation Particle Filtering
with Anticipated-Shock Quadrature and Piecewise Localization

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Abstract

We study likelihood evaluation for nonlinear DSGE models when the equilibrium transition is available only through finite-horizon extended-path solves and may contain kinks from occasionally binding constraints. The proposed approximation, pathwise perturbation particle filtering (PPPF), has four layers. First, at each filtering step we solve a finite-horizon stacked equilibrium system conditional on a low-dimensional auxiliary object of anticipated future shocks. Second, conditional on that auxiliary object and on a localization or regime index, we linearize the one-step mapping around the solved path, which yields an affine-Gaussian transition. Third, we integrate the auxiliary object with respect to its Gaussian reference measure by positive-weight quadrature, producing a Gaussian-mixture transition kernel. Fourth, we evaluate the resulting approximate likelihood with a Rao–Blackwellized particle filter that samples only the discrete mixture index and propagates the continuous state analytically.

The paper distinguishes the true DSGE transition from the finite-horizon stochastic perfect-foresight kernel, the local-linearization kernel, and the numerically integrated mixture kernel, so the source of bias is explicit at each step. We provide an end-to-end algorithm, a notation refactor, and a benchmark design that separates approximation bias from Monte Carlo variance. In a New Keynesian model with an effective lower bound, PPPF delivers substantial variance reduction relative to a bootstrap particle filter, but its bias is driven mainly by horizon truncation, coarse anticipated-shock integration, anchor reuse, and regime handling at the kink. In the benchmark specification, clustered anchors materially improve likelihood evaluation, while an ELB-aware discrete current-shock mixture repairs the direct impulse-response pathology that arose when the one-step innovation was closed with a single Gaussian covariance. Even after those fixes, however, sizable likelihood gaps remain at the long-sample benchmark, so the paper's conclusion is not that PPPF is solved, but that its remaining error is now concentrated in a much narrower and more interpretable part of the transition approximation.

Keywords: nonlinear DSGE estimation; particle filter; Rao–Blackwellization; extended path; stochastic extended path; dynamic perturbation; sigma points; occasionally binding constraints; effective lower bound.
JEL: C11, C15, C61, E32.

1 Introduction

Likelihood-based estimation of nonlinear DSGE models is hard for two related reasons. First, the transition law is rarely available in closed form: it is defined only implicitly by equilibrium conditions with conditional expectations, and those conditions may be approximated only numerically. Second, once that transition is embedded inside a particle filter, the likelihood estimator can have high Monte Carlo variance precisely in the regions where the model is most nonlinear or where occasionally binding constraints create kinks (Kantas et al., 2015; Andrieu et al., 2010).

The standard response is to compute a global or semi-global approximation to the policy functions and then filter the resulting state-space system. That strategy is often effective, but it is also demanding in medium-scale DSGEs and particularly delicate near occasionally binding constraints. The alternative studied here is local in state space but sequential in time. At each date, we solve a finite-horizon perfect-foresight problem along a path relevant for the current filtering distribution, linearize the one-step mapping around that path, and integrate a low-dimensional proxy for future uncertainty by numerical quadrature. The output is not a global decision rule. It is a one-step approximate transition kernel constructed for likelihood evaluation.

This perspective is close in spirit to extended path (Fair and Taylor, 1983), stochastic extended path (Adjemian and Juillard, 2025), and dynamic perturbation (Mennuni et al., 2025), but it serves a different purpose. Extended path and its stochastic variants are solution methods. Dynamic perturbation constructs local expansions along realized trajectories. PPPF embeds those ingredients inside a filtering recursion and therefore has to answer additional questions that solution methods can leave implicit: which objects are random and which are numerical integration nodes, when a quadrature rule may be interpreted as a probability mixture, how regime selection at a kink enters the likelihood recursion, and where approximation bias enters relative to Monte Carlo variance.

The main contribution of the paper is to separate those objects cleanly. The true DSGE transition kernel is denoted by K_θ. The finite-horizon stacked solve induces a second kernel that is already approximate because of horizon truncation and the terminal condition. Linearizing that solve around a reference path induces a third kernel, conditional on an auxiliary anticipated-shock variable and, when needed, on a localization or regime index. Integrating the auxiliary variable with respect to its Gaussian reference measure produces a Gaussian-mixture kernel. Only after those approximations are fixed do we evaluate the likelihood with a Rao–Blackwellized particle filter (RBPF). This decomposition matters because improvement in one layer need not reduce the dominant error in another.

The New Keynesian effective-lower-bound benchmark makes this point sharply. The benchmark shows that Rao–Blackwellization can reduce likelihood variance substantially, and that clustered anchors can reduce filtering bias further by limiting anchor reuse near the ELB. But the decisive transition-side gain comes from replacing the local Gaussian closure for the realized innovation with a discrete shock mixture in ELB-risk states. That change repairs the benchmark impulse responses far more than anchor localization does, while still leaving nontrivial long-sample likelihood gaps. A paper that wants to make a credible econometric claim therefore has to present those approximation layers explicitly rather than reporting a single PPPF likelihood as though it were a primitive object.

1.1 Contributions and organization

Section 2 positions PPPF relative to the particle-filter, extended-path, and occasionally binding constraint literatures. Section 3 introduces the state-space notation and resolves symbol collisions that would otherwise obscure the exposition. Sections 4–7 define the finite-horizon stacked system, derive the conditional affine-Gaussian transition, state the approximation taxonomy, and present the filtering recursion in a form that can be implemented. Sections 8 and 9 discuss which refinement arguments are valid, what they converge to, and when PPPF can dominate a global-solution-based likelihood evaluator under a fixed compute budget. Sections 10 and 11 contain the verification exercises and the New Keynesian ELB benchmark. The appendices collect the implicit-function derivation, the quadrature details, and benchmark-specific remarks.

2 Related literature

Likelihood-based estimation of DSGE models with nonlinear or non-Gaussian state dynamics typically relies on sequential Monte Carlo methods, often embedded inside parameter-estimation schemes (Kantas et al., 2015; Andrieu et al., 2010). In macroeconomic applications, particle-filter likelihoods have been used for estimation and model comparison in nonlinear DSGEs; see, e.g., Fernandez-Villaverde and Rubio-Ramirez (2007) and the textbook treatment in Herbst and Schorfheide (2016). Variance reduction and robustness near sharp nonlinearities have motivated methods such as tempered particle filtering (Herbst and Schorfheide, 2019) and auxiliary/proposal-based particle filters (Pitt and Shephard, 1999).

Occasionally binding constraint models introduce non-differentiabilities that complicate both solution and filtering. OccBin (Guerrieri and Iacoviello, 2015) and related finite-horizon regime-iteration methods produce piecewise-linear decision rules conditional on a binding pattern over a horizon, and have been widely used for ELB applications (Gust et al., 2017). Aruoba et al. (2021) develop piecewise-linear continuous (PLC) approximations for OBC DSGEs and a conditionally optimal particle filter (COPF) that exploits the resulting structure to reduce likelihood variance. PPPF is complementary rather than a replacement. It retains the filtering perspective of those papers but constructs the transition from pathwise solves and local Jacobians rather than from a global piecewise-linear policy map.

The extended-path method of Fair and Taylor (1983) solves a finite-horizon perfect-foresight system with a terminal condition, producing a time-varying local approximation to equilibrium dynamics that is often computationally attractive in medium-to-large models. Stochastic extended path (Adjemian and Juillard, 2025) improves on certainty equivalence by integrating over future shocks to better approximate conditional expectations. Dynamic perturbation (Mennuni et al., 2025) computes first-order expansions along simulated equilibrium paths and emphasizes accuracy along realized trajectories rather than at a fixed steady state. PPPF combines those ideas with Gaussian-sum filtering: conditional on a low-dimensional shock-path proxy and a regime index, one gets a locally affine state transition that can be propagated analytically inside an RBPF.

Gaussian-sum representations of nonlinear filtering problems date back to Alspach and Sorenson (1972). Rao–Blackwellized particle filters (Doucet et al., 2000) exploit conditional linear-Gaussian structure to integrate out continuous components analytically via Kalman filtering, reducing Monte Carlo variance relative to a full bootstrap filter (Gordon et al., 1993). Sigma-point quadrature and related deterministic integration rules are widely used to approximate Gaussian expectations and underlie unscented filtering (Julier and Uhlmann, 2004). PPPF uses such quadrature not as a single-Gaussian moment-matching device, but to produce an explicit mixture kernel that can be handled either by branching or by sampling mixture indices within an RBPF.

3 Nonlinear state-space model

3.1 Generic state-space formulation

Let $z_t \in \mathbb{R}^{n_z}$ denote the latent state used by the filter and let $y_t \in \mathbb{R}^{n_y}$ denote the data. The true transition kernel is $K_\theta(z_t, \mathrm{d}z_{t+1})$, where θ collects model parameters. To avoid symbol collisions later in the New Keynesian application, we reserve x_t for the output gap and use z_t for the generic latent state throughout the methodological sections.

The measurement density is written generically as

$$y_t \mid z_t \sim g_\theta(\cdot \mid z_t). \tag{1}$$

Symbol	Meaning
$z_t \in \mathbb{R}^{n_z}$	generic latent state used by the filter
$y_t \in \mathbb{R}^{n_y}$	observed data
$g_\theta(y_t \mid z_t)$	generic measurement density
$y_t = Mz_t + \eta_t$	linear-Gaussian measurement special case
$\eta_t \sim \mathcal{N}(0, R)$	measurement noise
$\varepsilon_{t+1} \sim \mathcal{N}(0, \Sigma)$	realized one-step innovation (random)
H	horizon of the finite-horizon stacked system
$\omega_t \sim \mathcal{N}(0, \Omega)$	auxiliary anticipated-shock proxy (numerical integration variable)
$\mu_\Omega(\mathrm{d}\omega)$	Gaussian measure $\mathcal{N}(0, \Omega)(\mathrm{d}\omega)$
$a_t \in \{1, \ldots, A_t\}$	deterministic anchor-group index used to localize path solves
$z_t^{\mathrm{ref},a}$	anchor/reference state for group a
$k_t \in \{1, \ldots, K\}$	quadrature node index attached to $\omega^{(k)}$
w_k	quadrature weights; if sampled they must be nonnegative and sum to one
$q_t \in \{1, \ldots, Q_t\}$	current-shock cubature index used only in ELB-risk states
ϖ_q	current-shock cubature weights; if sampled they must be nonnegative and sum to one
$s_t \in \mathcal{S}_t$	localization index: partition cell, regime path, or binding indicator
$j_t = (s_t, k_t, q_t)$	sampled mixture index conditional on anchor group a_t
x_t	output gap in the NK application only
π_t	inflation in the NK application only
i_t	policy rate in the NK application only

Table 1: Core notation.

The baseline filtering derivations use the linear-Gaussian special case

$$y_t = Mz_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, R), \tag{2}$$

because that is the setting in which the continuous state can be marginalized exactly by Kalman filtering. In the NK benchmark below, inflation and the output gap are measured linearly, while the observed policy rate is a lower-censored transformation of the latent state; that measurement is handled by a moment-projected Gaussian update and is therefore kept separate from the four core transition approximations.

3.2 Definitions and notation

Table 1 records the main random variables, numerical integration objects, and discrete indices. Every time we integrate out the anticipated-shock proxy ω_t, the integration is with respect to the Gaussian reference measure $\mu_\Omega(\mathrm{d}\omega) = \mathcal{N}(0, \Omega)(\mathrm{d}\omega)$.

3.3 Equilibrium transition and Gaussian shocks

The transition is induced by equilibrium conditions and structural innovations. For exposition we write the one-step law as

$$z_{t+1} = F(z_t, \varepsilon_{t+1}; \theta), \qquad \varepsilon_{t+1} \sim \mathcal{N}(0, \Sigma), \tag{3}$$

even though in DSGEs the map F is generally not available in closed form. The paper treats $K_\theta(z, \cdot)$ as the exact object of interest and regards PPPF as a sequence of controlled approximations to that kernel.

4 From stacked equilibrium conditions to a conditional linear-Gaussian law

This section derives the transition used by the filter. The key distinction is between random shocks and numerical integration devices. The realized innovation ε_{t+1} is random and moves the state from t to $t+1$. The anticipated-shock proxy ω_t is a numerical device used to approximate expectations beyond $t+1$. In the ELB benchmark there is a third object in ELB-risk states: a positive-weight node for the realized current innovation. That node is also a numerical integration device. It approximates the law of ε_{t+1} locally near the kink and is not an additional latent shock.

4.1 A finite-horizon "stochastic perfect foresight" system

Fix a horizon $H \geq 2$. Let

$$\omega_t = (\varepsilon^a_{t+2}, \varepsilon^a_{t+3}, \ldots, \varepsilon^a_{t+H-1}) \in \mathbb{R}^d, \qquad d = \max\{0, H-2\} n_\varepsilon,$$

collect a finite horizon of future innovations beyond the realized one-step shock. The law of ω_t is induced by the model's structural shocks,

$$\omega_t \sim \mathcal{N}(0, \Omega), \qquad \Omega = \operatorname{diag}(\Sigma, \ldots, \Sigma),$$

with $H-2$ blocks of Σ when $H > 2$. The variable ω_t is not an extra shock realization on top of ε_{t+1}. It is a numerical integration variable with respect to μ_Ω and is introduced only to approximate the conditional expectations inside the finite-horizon stacked system.

Let $U_t = (z_{t+1}, \ldots, z_{t+H-1})$ denote the unknown future path beyond the pinned initial state. For each admissible localization index $s_t \in \mathcal{S}_t$, define a stacked residual map

$$\Psi_H(U_t; z_t^{\mathrm{ref}}, \omega_t, s_t, \theta) = 0, \tag{4}$$

where s_t may represent a partition cell, a binding pattern, or another device that selects a smooth branch of the equilibrium system. Equation (4) includes equilibrium conditions from t through $t+H-2$ and a terminal condition at $t+H-1$. When s_t is fixed and the regime is smooth, (4) is an ordinary finite-horizon implicit system of the kind solved by extended-path methods. We denote a solution by $\bar{U}_t(z_t^{\mathrm{ref}}, \omega_t, s_t; \theta)$ and its first element by $\bar{z}_{t+1}(z_t^{\mathrm{ref}}, \omega_t, s_t; \theta)$.

4.2 Local perturbations around the path

The stacked system may be solved at a shared anchor, at a small bank of clustered anchors, or at each particle's current state. With a shared anchor, one uses a single reference state such as the weighted mean of the filtering distribution. With clustered anchors, one partitions the previous-period particle cloud into a small number of deterministic groups and solves at anchor states $z_t^{\mathrm{ref},a}$ attached to those groups. Per-particle anchors set $z_t^{\mathrm{ref},a} = z_t$. Moving from shared to clustered to per-particle anchors reduces extrapolation error but increases the number of stacked solves.

Fix an anchor $z_t^{\mathrm{ref},a}$, a future-shock proxy ω_t, and a regime index s_t, and let $\bar{U}_t$ solve (4). Linearizing the stacked residual map around that solution and differentiating with respect to the realized one-step innovation ε_{t+1} yields

$$\tilde{z}_{t+1} = A_t(z_t^{\mathrm{ref},a}, \omega_t, s_t)\, \tilde{z}_t + B_t(z_t^{\mathrm{ref},a}, \omega_t, s_t)\, \varepsilon_{t+1} + r_t^{\mathrm{lin}}, \tag{5}$$

where $\tilde{z}_t = z_t - z_t^{\mathrm{ref},a}$ and r_t^{lin} is the local Taylor remainder. Appendix A derives the Jacobian formulas by the implicit function theorem. Away from kinks, dropping the remainder and integrating the current innovation under its Gaussian law defines the PPPF one-step approximation conditional on (ω_t, s_t, a_t):

$$z_{t+1} \mid (z_t, \omega_t, s_t, a_t) \approx \mathcal{N}\Big(m_t^{(a_t,s_t)}(z_t, \omega_t),\, Q_t^{(a_t,s_t)}(\omega_t)\Big), \tag{6}$$

with

$$m_t^{(a,s)}(z_t, \omega_t) = \bar{z}_{t+1}(z_t^{\mathrm{ref},a}, \omega_t, s) + A_t(z_t^{\mathrm{ref},a}, \omega_t, s)\,(z_t - z_t^{\mathrm{ref},a}), \tag{7}$$

$$Q_t^{(a,s)}(\omega_t) = B_t(z_t^{\mathrm{ref},a}, \omega_t, s)\, \Sigma\, B_t(z_t^{\mathrm{ref},a}, \omega_t, s)^{\top}. \tag{8}$$

If the solve is performed per particle, then $z_t^{\mathrm{ref},a} = z_t$ and the affine correction term disappears. If shared or clustered anchors are used, (7) is a first-order extrapolation away from the relevant anchor.

Near the ELB kink, the benchmark specification does not always collapse the realized current innovation into the single covariance matrix (8). Instead, for anchors that are locally at risk of switching between slack and binding regimes, the law of the realized current innovation is approximated by a positive-weight cubature rule $\{(\varepsilon_{t+1}^{(q)}, \varpi_q)\}_{q=1}^{Q_t}$. Conditional on $(a_t, s_t, \omega_t, q_t)$, the method then linearizes around the path associated with the fixed current-shock node $\varepsilon_{t+1}^{(q_t)}$ and obtains a component mean $m_t^{(a,s,q)}(z_t, \omega_t)$ together with a residual covariance $Q_t^{(a,s,q),\mathrm{res}}$ that reflects only regularization or remaining smooth uncertainty. This current-shock mixture is the relevant refinement near the kink because a single Gaussian closure of ε_{t+1} can average across different ELB regimes and generate spurious impact responses.

Remark 1 (Relation to dynamic perturbation and SEP). Dynamic perturbation (Mennuni et al., 2025) also computes first-order expansions along equilibrium paths. SEP (Adjemian and Juillard, 2025) integrates over future shocks to better approximate conditional expectations than certainty equivalence. PPPF combines those ideas with a filtering recursion: the path solve and Jacobian calculation are solution steps, while the mixture interpretation and the RBPF are likelihood-evaluation steps.

5 Integrating out anticipated shocks

5.1 Gaussian-mixture transition kernel

For a fixed anchor group $a_t = a$ and a fixed localization rule or regime weight $\alpha_t^{(a)}(s \mid \omega)$, integrating out ω_t with respect to μ_Ω yields the approximate transition kernel

$$\widehat{K}_{\theta,H}^{\mathrm{pppf},a}(z_t,\, \mathrm{d}z_{t+1}) = \sum_{s \in \mathcal{S}_t} \int \alpha_t^{(a)}(s \mid \omega)\, \mathcal{N}\Big(\, \mathrm{d}z_{t+1};\, m_t^{(a,s)}(z_t, \omega), Q_t^{(a,s)}(\omega)\Big)\, \mu_\Omega(\, \mathrm{d}\omega). \tag{9}$$

Equation (9) is a Gaussian-sum kernel (Alspach and Sorenson, 1972). If localization is deterministic, then α_t is degenerate at one s. If regime uncertainty is carried explicitly, then α_t is a nonnegative mixing rule over regimes. In either case the integral is with respect to the Gaussian measure of the anticipated-shock proxy, not with respect to the realized shock ε_{t+1}.

In ELB-risk states, the benchmark specification replaces (9) by a mixed discrete-continuous version,

$$\widehat{K}^{\mathrm{pppf},a}_{\theta,H}(z_t,\, \mathrm{d}z_{t+1}) = \sum_{s\in\mathcal{S}_t} \int \sum_{q=1}^{Q_t} \alpha_t^{(a)}(s \mid \omega, q)\, \varpi_q\, \mathcal{N}\Big(\mathrm{d}z_{t+1};\, m_t^{(a,s,q)}(z_t,\omega), Q_t^{(a,s,q),\mathrm{res}}(\omega)\Big)\, \mu_\Omega(\mathrm{d}\omega), \tag{10}$$

where the sum over q approximates the law of the realized current innovation near the kink. Equation (10) still defines a nonnegative probability mixture when $\varpi_q \geq 0$ and the regime weights are nonnegative.

We approximate the integral numerically by a K-node rule:

$$\widehat{K}^{\mathrm{pppf},a}_{\theta,H,K}(z_t,\, \mathrm{d}z_{t+1}) = \sum_{k=1}^{K} \sum_{s\in\mathcal{S}_t} \sum_{q=1}^{Q_t} \tilde{w}_t^{(a,s,k,q)} \mathcal{N}\Big(\mathrm{d}z_{t+1};\, m_t^{(a,s,q)}(z_t,\omega^{(k)}), Q_t^{(a,s,q),\mathrm{res}}(\omega^{(k)})\Big), \tag{11}$$

where $\tilde{w}_t^{(a,s,k,q)} \geq 0$ and $\sum_{s,k,q} \tilde{w}_t^{(a,s,k,q)} = 1$ when the nodes are used as a probability mixture. In smooth regions the index q is degenerate and (11) reduces to the Gaussian-closure case. When the quadrature rule has signed weights, (11) is still a valid deterministic quadrature formula but not a sampling distribution.

5.2 Sigma points as deterministic "anticipated shock" draws

For $\omega \sim \mathcal{N}(0, \Omega) \in \mathbb{R}^d$, the basic unscented transform uses $2d+1$ sigma points and associated weights that match mean and covariance and are exact for polynomials up to degree three (for symmetric choices) (Julier and Uhlmann, 2004). Sigma points are attractive only when the anticipated-shock proxy is genuinely low dimensional. If d is large, positive-weight quadrature becomes too expensive for mixture filtering and the claimed variance reduction from Rao–Blackwellization is overwhelmed by component proliferation.

6 Discrete localization and "globalization" of the approximation

Quadrature refinement controls only the numerical integration error in (9) and (10). It does not repair bias from horizon truncation, terminal conditions, anchor reuse, or linearization across a kink. Those errors require separate choices about anchor localization, regime handling, and, near the ELB, the closure used for the realized current innovation.

6.1 Localization index: partition cells or regime/binding patterns

Two distinct localization problems matter. The first concerns anchors. If a single shared anchor is used, the local affine correction may be transported too far across the filtering distribution. Clustered anchors provide a middle ground: they partition the particle cloud into a small number of deterministic groups and attach one path solve and Jacobian block to each group. Per-particle anchors eliminate anchor-reuse error but can be prohibitively expensive.

The second problem concerns the regime or binding pattern of an occasionally binding constraint. This is the economically relevant case for ELB models. The correct object to differentiate

Approximation axis	Exact object being replaced	Computable surrogate	Main error mechanism
Finite horizon and terminal condition	Infinite-horizon rational-expectations equilibrium kernel K_θ	Kernel induced by the H-period stacked solve with a terminal condition	Truncation of future expectations and terminal misspecification
Expectation approximation	Finite-horizon kernel with exact integration over future shocks and exact local treatment of the current innovation law	Kernel based on low-dimensional proxy ω_t integrated with respect to μ_Ω, plus discrete current-shock cubature when activated near a kink	Omitted future uncertainty, poorly chosen shock proxy, or an inaccurate closure of the realized current innovation
Local linearization	Finite-horizon path solve conditional on (ω_t, s_t) or (ω_t, s_t, q_t)	Locally affine component (6) or its fixed-current-shock analogue near the ELB	Taylor remainder and, conditional on the chosen anchor, linearization error
Localization and regime handling	State-specific local reference point and regime law	Shared, clustered, or per-particle anchors together with explicit regime selection or regime mixtures over s_t	Anchor-reuse error, cell misclassification, or regime misclassification, especially near kinks and OBC boundaries

Table 2: Four core approximation axes in PPPF. Quadrature refinement changes only the expectation-approximation layer unless the localization scheme is also refined.

is not the max operator itself but the smooth equilibrium system conditional on a regime. OccBin (Guerrieri and Iacoviello, 2015) implements a hard regime choice. PPPF can instead carry a small bind/slack mixture near the boundary. Even that is not enough when the realized current innovation itself straddles the kink, which is why the benchmark introduces the discrete current-shock mixture in ELB-risk states.

6.2 Approximation taxonomy

Table 2 summarizes the four core approximation axes. The right comparison is always between neighboring layers in the construction, not between a crude PPPF likelihood and the true DSGE likelihood as if only one approximation had been made.

The NK benchmark holds fixed one additional approximation that is not part of the core PPPF transition design: the observed interest rate is updated with a moment-projected censored-Gaussian measurement rather than an exact non-Gaussian filter. That measurement approximation is held fixed across the PPPF ablations, so the reported differences are still informative about the transition-side errors summarized in Table 2.

7 Filtering and likelihood evaluation

7.1 PPPF-mixture-RBPF recursion

Conditional on an anchor group a_t, the approximate transition can be written as a discrete mixture of affine-Gaussian components:

$$z_{t+1} \mid (z_t, a_t, j_t = j) \approx A_t^{(j)} z_t + a_t^{(j)} + \xi_{t+1}^{(j)}, \qquad \xi_{t+1}^{(j)} \sim \mathcal{N}(0, Q_t^{(j)}).$$

With linear-Gaussian measurements the continuous state can be marginalized exactly conditional on the discrete history, yielding an RBPF (Doucet et al., 2000). With the censored policy-rate measurement used in the NK benchmark, the same recursion applies but the measurement update is only moment projected, not exact.

Each particle therefore stores a Gaussian state summary $(\mu_t^{(i)}, P_t^{(i)})$ and a weight $W_t^{(i)}$. Let $\ell_t^{(i)}(j)$ denote the predictive density of y_t under particle i and component j, after the Gaussian prediction step and before any resampling. The PPPF recursion is as follows.

1. Initialize N particles with $(\mu_0^{(i)}, P_0^{(i)}, W_0^{(i)})$, where $W_0^{(i)} = 1/N$.

2. For each date $t = 1, \dots, T$, construct anchor groups. In the shared-anchor version there is one group and

$$z_t^{\text{ref}} = \sum_{i=1}^{N} W_{t-1}^{(i)} \mu_{t-1}^{(i)}.$$

   In the clustered-anchor version there are groups $a = 1, \dots, A_t$ with anchors $z_t^{\text{ref},a}$ constructed from deterministic summaries of the previous-period particle cloud. In the per-particle version, each particle forms its own singleton group with anchor $z_t^{\text{ref},i} = \mu_{t-1}^{(i)}$.

3. For each required anchor group and each admissible index $j = (s, k, q)$, solve the stacked system (4) at $(z_t^{\text{ref},a}, \omega^{(k)}, s)$ and compute the corresponding component parameters. In smooth regions q is degenerate and the component covariance is the Gaussian closure (8). In ELB-risk regions, build a positive-weight discrete mixture over current-shock nodes and attach the resulting fixed-current-shock local components to the same anchor. Form the nonnegative anchor-specific mixture weights $\tilde{w}_t^{(a,j)}$. If signed quadrature is used, this step cannot feed a sampling-based RBPF and must instead be implemented by deterministic branching.

4. For each particle i, assign its deterministic anchor group $a(i)$ and, for each candidate component j in that group's mixture, run the Gaussian predict step

$$(\mu_{t|t-1}^{(i,j)}, P_{t|t-1}^{(i,j)}) = \big(A_t^{(j)} \mu_{t-1}^{(i)} + a_t^{(j)},\ A_t^{(j)} P_{t-1}^{(i)} A_t^{(j)\top} + Q_t^{(j)}\big),$$

   and evaluate the predictive density

$$\ell_t^{(i)}(j) = p(y_t \mid y_{1:t-1}, j, \mu_{t-1}^{(i)}, P_{t-1}^{(i)}).$$

5. Choose an index proposal $q_t^{(i)}(j)$. Two useful choices are the prior mixture $q_t^{(i)}(j) = \tilde{w}_t^{(a(i),j)}$ and the auxiliary proposal

$$q_t^{(i)}(j) \propto \tilde{w}_t^{(a(i),j)}\, \ell_t^{(i)}(j),$$

   which conditions on the current observation.

6. Sample $j_t^{(i)} \sim q_t^{(i)}(\cdot)$ and update the particle weight by

$$W_t^{(i)} \propto W_{t-1}^{(i)}\, \ell_t^{(i)}(j_t^{(i)})\, \frac{\tilde{w}_t^{(a(i),j_t^{(i)})}}{q_t^{(i)}(j_t^{(i)})}. \tag{12}$$

   Under the auxiliary proposal $q_t^{(i)}(j) \propto \tilde{w}_t^{(a(i),j)} \ell_t^{(i)}(j)$, the correction factor in (12) collapses to the normalizing constant $\sum_j \tilde{w}_t^{(a(i),j)} \ell_t^{(i)}(j)$.

7. Conditional on the sampled component, carry out the measurement update and obtain $(\mu_t^{(i)}, P_t^{(i)})$.
8. Normalize the weights, update the log-likelihood estimate by the log normalizing constant, compute ESS, and resample if ESS falls below a chosen threshold.

The computational bottleneck is the number of component evaluations per date, not the Kalman algebra itself. That is why the dimension of ω_t, the number of regimes carried near a kink, and the use of shared versus per-particle anchors are first-order design choices rather than incidental numerical details.

7.2 Mixture handling and quadrature choices

Mixture handling is the point at which many otherwise careful papers become ambiguous. Given a transition of the form $\sum_j \tilde{w}_t^{(j)} \mathcal{N}(m_t^{(j)}, Q_t^{(j)})$, moment-matching the mixture to a single Gaussian generally changes the likelihood. The PPPF object is the mixture itself. It must therefore be handled either by exhaustive branching or by sampling the discrete component and accounting for the proposal in (12).

This distinction is especially important when the quadrature rule comes from sigma points. Unscented-transform formulas are often presented as deterministic quadrature rules for moment calculation and may use negative weights. Such rules are admissible for deterministic integration but inadmissible for a sampling interpretation. If the paper speaks of "sampling quadrature nodes," then the corresponding weights must be nonnegative. The benchmark calculations enforce this restriction explicitly.

8 Approximation theory: kernel, filter, and likelihood

8.1 Quadrature refinement converges to the mixture kernel

Let $\widehat{K}_{\theta,H}^{\mathrm{pppf}}$ be the mixture kernel in (9) and let $\widehat{K}_{\theta,H,K}^{\mathrm{pppf}}$ be its quadrature approximation in (11).

Assumption 1 (Quadrature consistency)**.** *The quadrature rules $(\omega^{(k)}, w_k)_{k=1}^K$ are consistent for the Gaussian measure $\mathcal{N}(0, \Omega)$:*

$$\sum_{k=1}^{K} w_k h(\omega^{(k)}) \to \int h(\omega)\, \phi(\omega; 0, \Omega)\, \mathrm{d}\omega \quad \textit{for all bounded continuous } h.$$

Proposition 1 (Quadrature refinement)**.** *Assume that, for each fixed z and each fixed anchor group, the component means, component covariances, and nonnegative mixture weights entering (11) are continuous functions of the numerical integration variables and are dominated by an integrable envelope. Then, for every bounded continuous test function ψ,*

$$\int \psi(z')\, \widehat{K}_{\theta,H,K}^{\mathrm{pppf}}(z, \mathrm{d}z') \to \int \psi(z')\, \widehat{K}_{\theta,H}^{\mathrm{pppf}}(z, \mathrm{d}z').$$

Remark 2. This proposition is deliberately modest. It says only that numerical integration converges to the PPPF mixture kernel defined by the chosen horizon, localization rule, and linearization. It does not imply convergence to the true DSGE kernel K_θ.

8.2 Kernel error decomposition

The approximation taxonomy implies the one-step decomposition

$$\|K_\theta(z,\cdot) - \widehat{K}^{\text{pppf}}_{\theta,H,K,J}(z,\cdot)\| \le \underbrace{\|K_\theta - \widetilde{K}_{\theta,H}\|}_{\text{finite horizon and terminal condition}} + \underbrace{\|\widetilde{K}_{\theta,H} - K^{\omega}_{\theta,H}\|}_{\text{anticipated-shock proxy}} + \underbrace{\|K^{\omega}_{\theta,H} - \widehat{K}^{\text{pppf}}_{\theta,H}\|}_{\text{local linearization}}$$
$$+ \underbrace{\|\widehat{K}^{\text{pppf}}_{\theta,H} - \widehat{K}^{\text{pppf}}_{\theta,H,K}\|}_{\text{quadrature}} + \underbrace{\|\widehat{K}^{\text{pppf}}_{\theta,H,K} - \widehat{K}^{\text{pppf}}_{\theta,H,K,J}\|}_{\text{localization or regime handling}} . \qquad (13)$$

Here $\widetilde{K}_{\theta,H}$ denotes the kernel induced by the exact H-period stacked problem, $K^{\omega}_{\theta,H}$ is the same kernel after replacing the full future uncertainty with the low-dimensional proxy ω_t, and $\widehat{K}^{\text{pppf}}_{\theta,H,K,J}$ is the final computable kernel. Equation (13) is the bookkeeping device that the benchmark will use to interpret the ablations.

8.3 From kernel error to filter and likelihood error

Let $\delta_t = \sup_z \|K_\theta(z,\cdot) - \widehat{K}^{\text{pppf}}_{\theta,H,K,J}(z,\cdot)\|_{\text{TV}}$ denote the one-step kernel error. Under standard hidden-Markov stability conditions, perturbation bounds imply that filtering and log-likelihood errors inherit the same structure up to forgetting factors; see Del Moral (2004); Whiteley (2013); Kantas et al. (2015).

Assumption 2 (Uniform mixing and bounded likelihood)**.** *There exists $\epsilon \in (0,1)$ and a probability measure ν on $\mathcal{X}$ such that for all $x \in \mathcal{X}$ and measurable A,*

$$K_\theta(x, A) \ge \epsilon \nu(A),$$

and the measurement density satisfies $0 < g_{\min} \le g_\theta(y \mid x) \le g_{\max} < \infty$ for all $x \in \mathcal{X}$ and realized y_t.

Remark 3 (Strength of the global minorization condition). Assumption 2 is a standard sufficient condition in stability theory, but it is too strong for many DSGEs. It is included only to justify the qualitative message that reducing the dominant one-step error in (13) is the relevant route to improving the likelihood. The benchmark evidence below should therefore be read as diagnostic rather than as a proof of asymptotic superiority.

Proposition 2 (Standard perturbation implication)**.** *Under Assumption 2, there exist constants $C < \infty$ and $\rho \in (0,1)$ such that*

$$\|\pi_{\theta,t} - \widehat{\pi}_{\theta,t}\|_{\text{TV}} \le C \sum_{s=1}^{t} \rho^{t-s} \delta_s, \qquad t \ge 1.$$

An analogous bound holds for the cumulative log-likelihood error with $\sum_{t=1}^{T} \delta_t$ on the right-hand side, up to the same forgetting factors.

9 When can PPPF be "better" than a global solution?

Let ℓ_θ denote the true log-likelihood and let $\widehat{\ell}^{(A)}_\theta$ be a particle-based estimator using approximation strategy $A \in \{\text{glob}, \text{pppf}\}$. Under a compute budget C, the feasible particle count is roughly

$N_A \approx C/c_A$, where c_A is cost per particle per period. A standard decomposition is

$$\mathbb{E}\left[(\widehat{\ell}_\theta^{(A)} - \ell_\theta)^2\right] = \underbrace{\left(\mathrm{Bias}_A\right)^2}_{\text{approximation bias}} + \underbrace{\frac{\sigma_A^2}{N_A}}_{\text{Monte Carlo variance}}.$$

Rao–Blackwellization reduces σ^2_{pppf} by integrating out part of the latent state analytically (Doucet et al., 2000). A sufficient dominance condition is therefore

$$(\mathrm{Bias}_{\mathrm{pppf}})^2 + \sigma^2_{\mathrm{pppf}} \frac{c_{\mathrm{pppf}}}{C} < (\mathrm{Bias}_{\mathrm{glob}})^2 + \sigma^2_{\mathrm{glob}} \frac{c_{\mathrm{glob}}}{C}.$$

This inequality is not a theorem about PPPF in general. It is the practical criterion that the benchmark is designed to illuminate.

10 A minimal toy example (no DSGE background)

The NK–ELB benchmark in Section 11 combines all four approximation axes. The present example strips away the DSGE structure and isolates only the quadrature-as-mixture logic.

Consider the one-dimensional state-space model

$$y_t = x_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, r),$$

and

$$x_{t+1} \mid (x_t, \omega_t) = \rho x_t + b \tanh(\omega_t) + \sigma \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0,1), \qquad \omega_t \sim \mathcal{N}(0, \Omega).$$

Integrating out ω_t with respect to $\mathcal{N}(0, \Omega)$ yields a Gaussian-mixture transition of the form (9). Certainty equivalence corresponds to the degenerate choice $\omega_t \equiv 0$. The example is useful because a high-accuracy likelihood can be computed by fine quadrature or grid discretization, so one can see directly that increasing K reduces expectation-approximation error, while collapsing the mixture to a single Gaussian alters the likelihood even if the first two moments are preserved.

11 Benchmark: a New Keynesian model with an ELB

This section specifies the benchmark environment used for the numerical comparison. The benchmark is intentionally narrow: it is designed to diagnose where PPPF bias enters in a simple ELB environment, not to establish dominance over every alternative solution method.

11.1 Core equilibrium conditions

We consider a canonical three-equation New Keynesian model with an effective lower bound (ELB) on the nominal rate. Let x_t denote the output gap, π_t inflation, and i_t the nominal interest rate. A standard log-linearized NK system is

$$x_t = \mathbb{E}_t x_{t+1} - \sigma^{-1}\Big(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n\Big), \tag{14}$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t, \tag{15}$$

$$i_t^\star = \phi_\pi \pi_t + \phi_x x_t + \nu_t + i^{\mathrm{ss}}, \tag{16}$$

$$i_t = \max\{\underline{i},\ i_t^\star\}. \tag{17}$$

The exogenous processes are Gaussian AR(1) laws for r_t^n and ν_t. The ELB condition (17) creates the kink that drives the regime problem.

11.2 State-space representation and measurement

A minimal latent state is

$$z_t = (x_t, \pi_t, r_t^n, \nu_t)^\top.$$

The policy rate is not stored as an independent state variable; it is an algebraic function of z_t through (16)–(17). The observable vector is

$$y_t = (\pi_t^{\text{obs}}, x_t^{\text{obs}}, i_t^{\text{obs}})^\top.$$

Inflation and the output gap are measured linearly with Gaussian noise. The policy rate is measured as a lower-censored function of z_t with Gaussian observation error. Consequently, the transition-side RBPF is exact conditional on the discrete index only for the first two measurements; the third measurement is handled by a Gaussian moment projection in the benchmark calculations.

11.3 Competing methods

The benchmark compares three likelihood-evaluation strategies that differ in how they approximate the transition kernel near the ELB kink and how they control Monte Carlo variance. First, an OccBin-style piecewise-linear extended-path solution (Guerrieri and Iacoviello, 2015) is embedded in a bootstrap particle filter. Second, the benchmark also considers a PLC-inspired baseline that interpolates the global discretized benchmark and filters only the exogenous state with an observation-informed Gaussian proposal. This is best interpreted as a PLC/COPF-style comparator, not as a line-by-line reimplementation of Aruoba et al. (2021). Third, PPPF constructs an anchor-specific Gaussian-mixture transition by integrating over a short-horizon anticipated-shock proxy and, in ELB-risk states, by replacing the single-Gaussian closure of the realized current innovation with a small positive-weight discrete mixture.

11.4 Proposed evaluation metrics

Because particle filters yield stochastic likelihood approximations, the comparison emphasizes runtime per likelihood evaluation, Monte Carlo variability of repeated log-likelihood estimates on fixed data, ESS profiles (and resampling frequency), and impulse responses relative to the discretized global benchmark. The aim is to distinguish approximation bias from Monte Carlo variance. Mean log-likelihood gaps and IRF discrepancies are interpreted as kernel differences. Dispersion across repeated runs and ESS are interpreted as Monte Carlo diagnostics.

11.5 Experimental design

The empirical design combines an analytic verification exercise (Burnside recursion) for the expectation approximation component with a New Keynesian ELB benchmark. The Burnside exercise isolates the value of integrating over a curved expectation term. The no-ELB verification isolates the filtering layer. The ELB benchmark then combines horizon truncation, expectation approximation, local linearization, and regime handling.

In the NK–ELB experiment, the OccBin extended-path horizon is $H = 20$. We solve the discretized global benchmark (Tauchen grid) and simulate a dataset of length $T = 250$ from that benchmark, adding small independent measurement noise on (π_t, x_t, i_t). For likelihood evaluation we use $N = 256$ particles and repeat each filter run $N_{\text{rep}} = 30$ times with different random seeds to measure Monte Carlo variability.

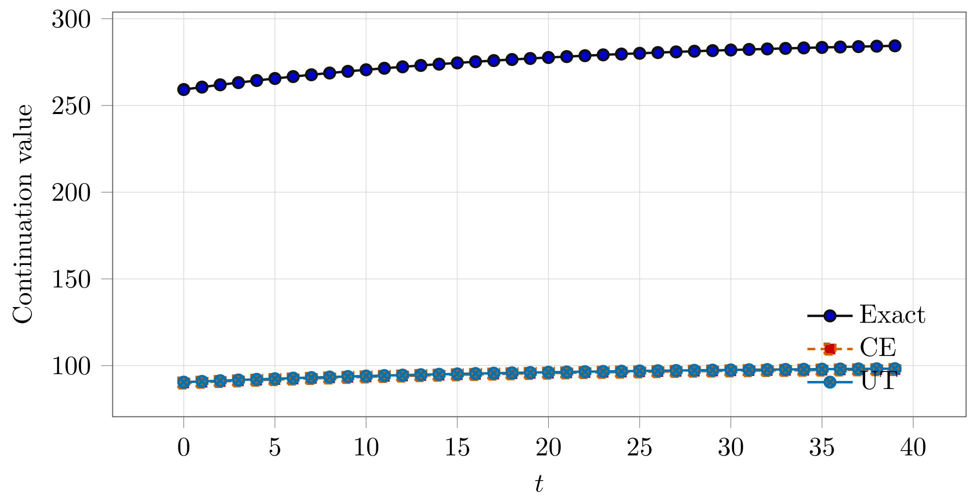

Figure 1: Burnside recursion verification exercise: exact benchmark versus certainty equivalence (CE) and unscented-transform (UT) quadrature.

To attribute PPPF likelihood bias in Table 4 to specific approximation steps, the ablations vary the quadrature resolution K (including $K = 1$ certainty equivalence), the horizon and terminal condition, the degree of anchor reuse, and the regime treatment at the ELB boundary. The direct IRF diagnostic in Figure 2 separately evaluates the one-step current-shock closure used near ELB-risk states. Reporting the mean log-likelihood gap to the discretized benchmark under these perturbations turns the bias–variance tradeoff into an explicit bias attribution exercise.

The discretized global benchmark induces a finite-state hidden Markov model for the exogenous state. Let $g_t \in \{1, \ldots, S\}$ index the Tauchen grid point for (r_t^n, ν_t) and let $P \in \mathbb{R}^{S\times S}$ be the transition matrix. Solving the discretized model gives an expectation-consistent observable map $h(g) = (\pi(g), x(g), i(g))^\top$. With Gaussian measurement noise, the likelihood of the discretized benchmark is obtained exactly by the HMM forward recursion

$$\alpha_{t|t-1}(g') = \sum_{g=1}^{S} \alpha_{t-1}(g)\, P(g, g'), \qquad \tilde{\alpha}_t(g) = \alpha_{t|t-1}(g)\, p(y_t \mid g), \qquad z_t = \sum_{g=1}^{S} \tilde{\alpha}_t(g),$$

with $\alpha_t(g) = \tilde{\alpha}_t(g)/z_t$ and $\log p(y_{1:T}) = \sum_{t=1}^{T} \log z_t$. This likelihood is exact for the discretized benchmark, not for the underlying continuous-state DSGE. That distinction is essential: the benchmark is a high-quality comparison object, not literal truth.

11.6 Illustrative results

We begin with an analytic Jensen-style validation based on Burnside's recursion, where an exact benchmark is available. Figure 1 reports impulse responses for the continuation value under the exact benchmark, a certainty-equivalence approximation, and a sigma-point (UT) one-step quadrature approximation. The UT approximation tracks curvature effects better than certainty equivalence, illustrating why explicit integration over a low-dimensional shock object can matter even when the underlying state is Gaussian.

As an additional verification exercise in a setting where all methods should agree, we validate the filtering layer on a one-dimensional linear-Gaussian state-space model with known Kalman-filter likelihood. In that case, an RBPF with a single mixture component ($K = 1$) coincides with

Method	Mean loglik	SD loglik	Mean runtime (ms)	SD runtime (ms)	Mean ESS
Kalman (exact; no ELB)	2345.0	0.0	0.0	0.0	–
OccBin-PF (no ELB)	2342.8	1.8	745.0	10.0	196.3
PPPF-RBPF (no ELB)	2345.0	0.0	1226.6	8.9	2048.0

Table 3: NK no-ELB verification exercise: with the ELB inactive the model is linear-Gaussian, and PPPF with $K = 1$ and $\mathrm{Var}(\omega) = 0$ coincides with the Kalman likelihood. Statistics are reported across $N_{\text{rep}} = 10$ repeated filter runs on fixed data, with $T = 250$ and $N = 2048$ particles.

Method	Mean loglik	SD loglik	Mean time (ms)	SD time (ms)	Mean ESS
Benchmark (exact)	2495.0	0.0	13.3	0.0	–
OccBin-PF	2199.4	8.9	173.6	7.2	27.6
PPPF-RBPF	2120.5	7.1	1194.2	70.7	161.1
PLC-COPF	2405.6	1.3	30.5	2.7	160.8

Table 4: NK–ELB benchmark: repeated log-likelihood estimates on fixed data across $N_{\text{rep}} = 30$ filter runs, with $T = 250$ and $N = 256$ particles. The benchmark likelihood is exact for the discretized model.

the Kalman filter and matches the exact log-likelihood to machine precision. A bootstrap particle filter, which replaces analytic Gaussian integration with Monte Carlo integration, approaches the same likelihood as the particle count increases.

To verify that the same logic applies in the New Keynesian environment, we rerun the NK model with the ELB effectively removed (setting $\underline{i}$ far below any realized policy rate so the constraint never binds) and simulate data from the resulting linear-Gaussian state-space system driven by continuous AR(1) shocks. In this case an exact likelihood is available by Kalman filtering on the exogenous state under the implied linear policy mapping. The "exact" baseline is exact for the same approximate no-ELB model that the filters evaluate, so any discrepancy is purely a filtering error. Table 3 reports that PPPF (degenerating to a single component by setting $\mathrm{Var}(\omega) = 0$) matches the Kalman likelihood to numerical precision, while OccBin-PF converges to the same likelihood up to Monte Carlo error.

We then turn to the NK–ELB benchmark. In what follows, OccBin-PF denotes OccBin coupled with a bootstrap particle filter, PPPF-RBPF denotes the PPPF mixture kernel coupled with a Rao–Blackwellized particle filter, and PLC-COPF denotes the PLC-inspired comparator described above. Table 4 reports summary statistics from the benchmark comparison. We simulate data from the discretized global benchmark and compute its exact likelihood by the HMM forward algorithm described above. Relative to OccBin-PF, PPPF-RBPF delivers much higher ESS and slightly lower cross-run dispersion of the likelihood estimator, which is consistent with the variance-reduction role of Rao–Blackwellization. Its mean long-sample log-likelihood nevertheless remains well below the exact discretized benchmark, and in this calibration it also remains below OccBin-PF. The main message of Table 4 is therefore variance reduction without commensurate transition accuracy.

To diagnose which approximation components drive PPPF bias in this calibration, Table 5 reports a one-factor-at-a-time ablation suite holding fixed the simulated dataset and (except where noted) the particle count and number of repeated filter runs. The reported gap is the PPPF log-likelihood minus the exact benchmark likelihood for the discretized model, so it isolates differences in the approximate transition kernel rather than Monte Carlo variability of the particle filter. Several facts stand out. Moving from a single shared anchor to clustered anchors materially improves the likelihood, which confirms that anchor reuse is quantitatively important near the ELB. By

Case	H	ω horizon	K	Gap	SD	Mean ESS
Baseline	20	1	3	-373.4	6.4	163.0
Shared anchor	20	1	3	-416.0	18.2	158.8
CE (K=1)	20	1	1	-419.0	8.6	209.0
ω horizon 2	20	2	5	-442.3	4.9	156.7
ω horizon 3	20	3	7	-520.8	7.4	156.0
H=10	10	1	3	-559.9	4.2	170.0
H=40	40	1	3	-376.1	5.8	164.8
Fixed bind0	20	1	3	-414.4	21.2	161.9
Bind0 mixture	20	1	6	-378.2	6.0	165.4
Linear i	20	1	3	-388.4	8.2	164.3
No aux proposal	20	1	3	-399.6	14.3	85.2
Per-particle anchor†	20	1	3	-148.3	13.2	3.0

Table 5: NK–ELB PPPF ablations: mean log-likelihood gap to the exact discretized benchmark, standard deviation across repeated filter runs, and mean ESS. All rows use $T = 250$, $N = 256$, and $N_{\text{rep}} = 30$ unless marked with †, which indicates a reduced configuration ($T = 50$, $N = 4$, $N_{\text{rep}} = 2$) used only to illustrate the effect of per-particle anchoring.

contrast, certainty equivalence, deeper anticipated-shock horizons with the same local closure, and a shorter horizon all worsen the gap. Fixing the first-period binding decision across nodes is especially costly, while allowing a small bind/slack mixture largely reverses that deterioration. The table therefore points to anchor localization and ELB regime handling as the main determinants of the long-sample PPPF likelihood in this benchmark.

Figure 2 is best read as a diagnostic for the direct transition approximation near the ELB. Under the ELB-risk current-shock mixture described above, PPPF no longer exhibits the most obvious impact-response pathology generated by a single-Gaussian closure of that shock. The resulting PPPF IRF is still noticeably more contractionary than the discretized global benchmark and remains less accurate than the OccBin path in this calibration. Taken together with Table 4, this implies that the remaining PPPF likelihood gap reflects transition-side approximation error rather than particle-filter noise.

12 Conclusion

PPPF should be understood as a structured approximation scheme for likelihood evaluation, not as a single approximation. Its output depends on a horizon choice, a proxy for future uncertainty, a local-linearization rule, and a localization or regime device. Once those layers are stated explicitly, the role of the RBPF becomes clear: it reduces Monte Carlo variance conditional on the approximate transition, but it does not remove transition-side bias.

The ELB benchmark shows both the promise and the limitation of the approach. Rao–Blackwellization raises ESS sharply, clustered anchors improve likelihood relative to a single shared anchor, and an ELB-aware discrete current-shock mixture repairs the most obvious direct-IRF pathology created by a single-Gaussian closure of the realized innovation. But those gains do not eliminate the main quantitative problem: in the long-sample benchmark PPPF remains far from the discretized global likelihood. The most credible path forward is therefore further work on the transition approximation near the ELB rather than a claim that PPPF is already ready for structural estimation.

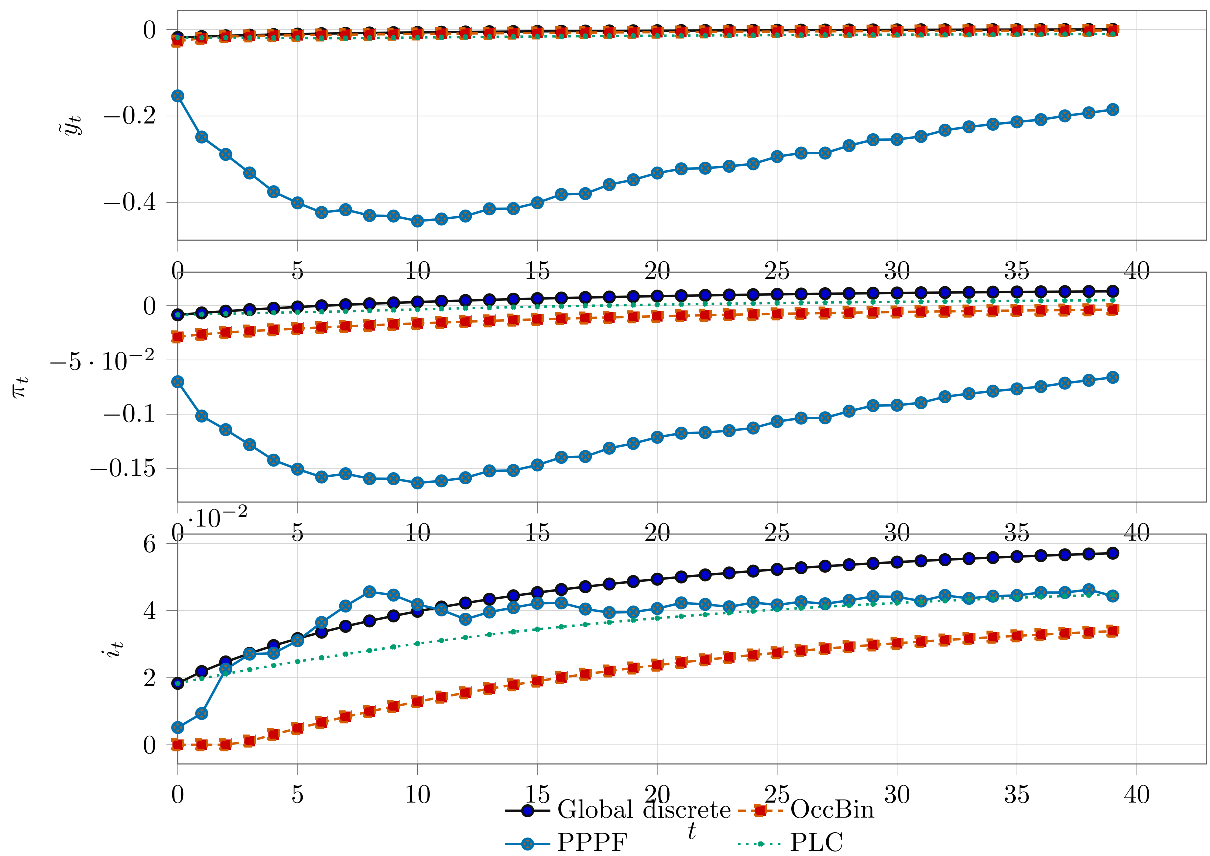

Figure 2: NK–ELB impulse responses: discretized global benchmark versus OccBin and PPPF-mixture approximations.

A Deriving the conditional linear system via the implicit function theorem

This appendix derives (5) from the stacked system. Let

$$U_t = (z_{t+1}, z_{t+2}, \ldots, z_{t+H-1}) \in \mathbb{R}^{(H-1)n_z}.$$

Write the regime-conditioned stacked system as

$$\Psi_H(U_t; z_t, \omega_t, s_t, \theta) = 0.$$

Suppose $\bar{U}_t$ solves this system for fixed (z_t, ω_t, s_t) and assume that the Jacobian with respect to U_t is nonsingular:

$$D_U \Psi_H(\bar{U}_t; z_t, \omega_t, s_t, \theta) \text{ is invertible.}$$

Then the implicit function theorem yields a local map $U_t = G(z_t, \omega_t, s_t)$ with derivatives

$$D_z G = -\big(D_U \Psi_H\big)^{-1} D_z \Psi_H, \qquad D_\omega G = -\big(D_U \Psi_H\big)^{-1} D_\omega \Psi_H.$$

To incorporate the realized innovation ε_{t+1}, augment the period-$t+1$ equations by the contemporaneous shock terms and differentiate with respect to ε_{t+1}. The first block row of $D_z G$ gives A_t and the first block row of $D_\varepsilon G$ gives B_t. Because the stacked Jacobian is sparse and banded, these derivatives are obtained by solving sparse linear systems rather than by differentiating a global policy map.

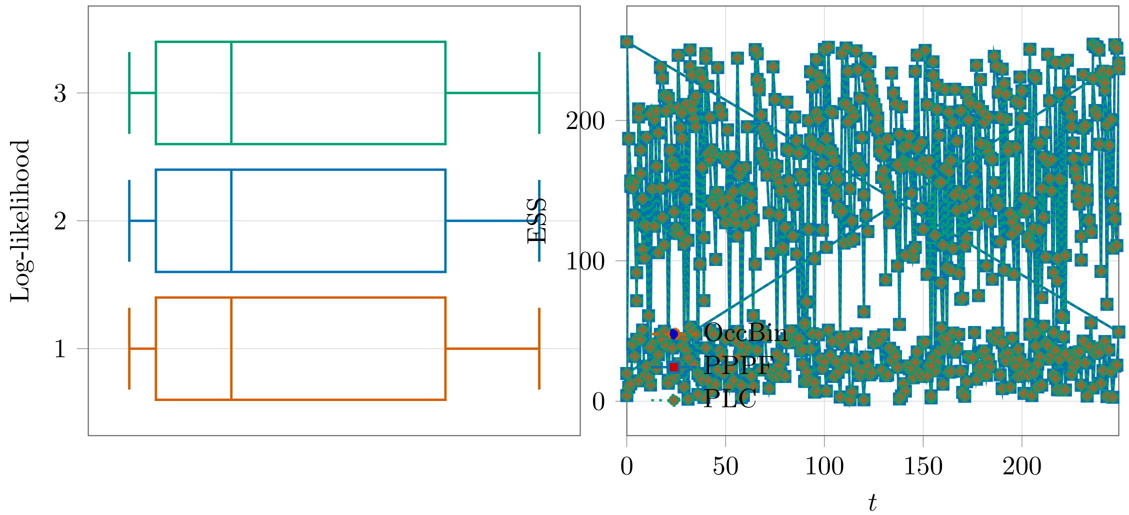

Figure 3: NK–ELB benchmark diagnostics: distribution of repeated log-likelihood estimates (boxplots) and ESS profile over time.

B Unscented-transform sigma points

For $\omega \sim \mathcal{N}(0, \Omega) \in \mathbb{R}^d$, the basic sigma-point set uses $2d + 1$ points:

$$\omega^{(0)} = 0, \qquad \omega^{(i)} = +\left[\sqrt{(d+\lambda)\Omega}\right]_{\cdot i}, \qquad \omega^{(i+d)} = -\left[\sqrt{(d+\lambda)\Omega}\right]_{\cdot i}, \quad i = 1, \ldots, d,$$

with weights

$$w_0^m = \frac{\lambda}{d+\lambda}, \qquad w_i^m = \frac{1}{2(d+\lambda)} \ (i \geq 1),$$

and covariance weights $w_0^c = w_0^m + (1 - \alpha^2 + \beta)$, $w_i^c = w_i^m$. Standard choices are $\alpha \in (0, 1]$, $\beta = 2$ (Gaussian prior), $\kappa = 0$ (Julier and Uhlmann, 2004). If sigma points are used as a probability mixture, the relevant weights are the mixture weights, not the covariance weights, and they must satisfy $w_k \geq 0$. For the UT parameterization above, this requires $\lambda \geq 0$. The common small-α settings used in unscented Kalman filtering typically violate this restriction and should therefore be interpreted as signed quadrature rather than as a sampling distribution. When nonnegative weights are required, a simple positive-weight rule for $\omega \sim \mathcal{N}(0, \Omega)$ uses $2d$ points:

$$\omega^{(i)} = +\sqrt{d}\,\left[\sqrt{\Omega}\right]_{\cdot i}, \qquad \omega^{(i+d)} = -\sqrt{d}\,\left[\sqrt{\Omega}\right]_{\cdot i}, \quad i = 1, \ldots, d, \qquad w_i = \frac{1}{2d}.$$

This yields a bona fide probability mixture and is exact for Gaussian polynomials up to degree three.

C Microfoundation and variants of the NK–ELB benchmark

Equations (14)–(15) arise from household Euler equations and Calvo price setting under standard assumptions; see Woodford (2003) for a textbook derivation. The ELB (17) is the occasionally binding constraint that motivates the regime index. In the benchmark specification the latent state is $z_t = (x_t, \pi_t, r_t^n, \nu_t)^\top$ and the observed policy rate is obtained from (16)–(17) rather than treated as an independent state variable.

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